**Project Journal - Milestone 2: Abusive Language Detection - Jaime Morales.**

Topic: Abusive Language Detection in Social Networks

**1. Software Architecture**

The system is implemented as a flat, three-stage pipeline. Each stage is a standalone script that can be executed independently.

- **src/preprocess.py:** Loads the raw Jigsaw CSV, derives a binary label from the six toxicity columns, balances the dataset to 1,500 toxic and 1,500 clean samples, applies text cleaning (lowercasing, URL removal, special character removal), and splits the result into train (2,100), validation (300), and test (600) sets saved to data/processed/.
- **src/train.py:** Loads the processed splits, tokenizes the text using the RoBERTa tokenizer, fine-tunes RoBERTa-base for sequence classification using PyTorch, and saves the trained model weights to models/roberta-jigsaw/.
- **src/evaluate.py:** Loads the saved model, runs inference on the test set, and outputs accuracy, F1-score, a full classification report, and a confusion matrix saved to results/.

Data flows linearly: data/raw/ → data/processed/ → models/ → results/. No APIs, no abstraction layers, and no intermediate databases are used.

**2. Technologies and Algorithms**

The following technologies were selected for the implementation, each with a justification and at least one alternative.

**Primary Model: RoBERTa-base**

RoBERTa (Robustly Optimized BERT Pretraining Approach) is a transformer encoder pre-trained via Masked Language Modeling (MLM). It was chosen because its bidirectional attention captures context from both sides of a token, which is essential for detecting subtle or implicit abuse. The roberta-base variant (125M parameters) is manageable on CPU for small datasets.

**Alternative: BERT-base.** BERT is the predecessor of RoBERTa and shares the same transformer architecture. The main differences are that RoBERTa uses dynamic masking, removes the Next Sentence Prediction objective, and was trained on significantly more data (~160GB vs ~16GB). In practice, RoBERTa consistently outperforms BERT on downstream NLP tasks. The disadvantage is that RoBERTa is marginally heavier in terms of training time. For this project, RoBERTa is preferred due to its stronger generalization, which is directly relevant to the domain transfer objective.

**Dataset: Jigsaw Toxic Comment Classification**

The Jigsaw dataset contains approximately 160,000 comments from Wikipedia talk pages, labeled across six toxicity categories. A binary label is derived: a comment is toxic if any category is positive. A balanced subset of 3,000 samples was extracted to avoid class imbalance issues during fine-tuning.

**Alternative: OLID (Offensive Language Identification Dataset).** Used in SemEval-2019, OLID provides a hierarchical label system (offensive vs. non-offensive, targeted vs. untargeted). Its advantage is a more fine-grained taxonomy of abuse. Its disadvantage is that it is Twitter-based, which would reduce the contrast between source and target domains needed for the domain transfer experiments. Jigsaw was chosen precisely because it comes from a different platform (news comment sections), making the domain gap more meaningful.

**Framework: PyTorch and Hugging Face Transformers**

PyTorch provides the tensor computation and backpropagation engine. Hugging Face Transformers provides the pre-trained RoBERTa model and tokenizer. The AdamW optimizer with a learning rate of 2e-5 is used, which is the standard setting for fine-tuning transformer models.

**Alternative: TensorFlow / Keras.** TensorFlow offers equivalent functionality and also integrates with Hugging Face. PyTorch was preferred for its more flexible and Pythonic API, its dominant adoption in NLP research, and its better native support for custom training loops, which gives more control over the fine-tuning process.

### 3. SSL-Related Focus

The Self-Supervised Learning component of this project is RoBERTa's pre-training via Masked Language Modeling (MLM). In MLM, 15% of input tokens are randomly masked and the model must predict the original tokens using the surrounding context. No human-labeled data is required: the supervision signal comes from the text itself, which is what makes it self-supervised.

This pre-training phase gives RoBERTa its general language understanding before any fine-tuning. When fine-tuned on the Jigsaw dataset, the model already understands grammar, semantics, and word relationships. The fine-tuning step only adapts the top classification layer to the specific task.

The connection to domain transfer is direct. A model pre-trained on general web text (SSL) and fine-tuned on one domain may fail on a different domain because the linguistic patterns, slang, and norms differ. This degradation across domains is the central research question of this project.

### 4. Current Stage of Implementation and Beta Results

All three pipeline scripts have been implemented and executed. The model was fine-tuned for one epoch on the training subset of 2,100 samples, running entirely on CPU (Intel Iris Xe, no NVIDIA GPU available). The reduced dataset size and single epoch were chosen to produce results within the hardware constraints of this stage.

Beta results after one epoch of fine-tuning (evaluated on the validation set of 300 samples):

- Training Loss: 0.3166
- Validation Accuracy: 0.9433
- Validation F1-Score: 0.9420

Test set results (600 samples, held-out):

- Test Accuracy: 0.9067
- Test F1-Score: 0.9082
- Clean class — Precision: 0.92 | Recall: 0.89 | F1: 0.91
- Toxic class — Precision: 0.89 | Recall: 0.92 | F1: 0.91

The high accuracy and F1 on a balanced validation set indicate that RoBERTa's pre-trained representations transfer effectively to toxicity classification even with minimal fine-tuning. These are preliminary results. Full evaluation on the held-out test set of 600 samples, including a classification report and confusion matrix, will be completed before the presentation.

**5. Next Steps**

- Complete evaluation on the test set and collect the full classification report and confusion matrix.
- Obtain a second dataset from a different domain (e.g., Twitter / OLID) to perform the domain transfer experiment.
- Fine-tune a second model instance on the new domain and compare performance metrics across domains.
- Analyze which types of abusive language degrade most under domain shift, connecting back to the typology proposed by Waseem et al. (2017).